BERDASARKAN KISI-KISI

Spesifikasi Teknis & Kebutuhan Aplikasi

A. Struktur Database (db_pemesanan_makanan)

Anda diminta membuat 3 tabel utama menggunakan fitur Migration:

foods (Master Makanan): id, name, category (Enum: Makanan, Minuman, Cemilan), price, description, image, timestamps.

orders (Transaksi Pesanan): id, customer_name, table_number, total_price, status (Enum: Pending, Diproses, Selesai), timestamps.

order_details (Detail Transaksi): id, order_id (Foreign Key), food_id (Foreign Key), quantity, subtotal, timestamps.

Peran & Alur Kerja Aplikasi (Workflow)

Sisi Customer (Publik):

Melihat daftar menu makanan berdasarkan kategori.

Memilih makanan, menginput jumlah item, mengisi nama & nomor meja, lalu melakukan checkout pesanan.

Sisi Admin (Terproteksi/Login):

Melihat rekap pesanan yang masuk.

Mengelola data makanan melalui modul CRUD lengkap (Tambah, Lihat, Update, Hapus).

Instruksi Kerja & Komponen Kunci Penilaian

Langkah / Unit	Instruksi Utama
1. Instalasi	Menyiapkan XAMPP/Laragon, PHP 8.x, Composer, Node.js, VS Code. Menginstal Laravel baru dan konfigurasi .env.
2. Struktur Data	Membuat Migration (3 tabel + Relasi FK) dan Seeder (minimal 5 data dummy makanan).
3. Logika Bisnis	Menerapkan MVC, memisahkan routes Admin & Customer, menghitung total harga, dan menangani transaksi multi-tabel (DB::transaction).
4. UI/UX	Membuat layout responsif, form input CRUD (termasuk upload gambar & dropdown), serta rekap data dalam bentuk tabel rapi.
5. Library/Package	Memakai Framework CSS (Bootstrap/Tailwind) & Package Laravel (Laravel Breeze/Fortify atau fitur Validator/File upload).
6. Best Practices	Mengikuti standar penulisan PSR (StudlyCaps, camelCase), memasang proteksi @csrf, dan enkripsi password.
7. Debugging	Pengujian fungsionalitas hingga bebas bug/error (Zero-Error Code).
8. Dokumentasi	Menambahkan komentar pada method Controller (PHPDoc/Inline) & membuat file

README.md (petunjuk instalasi, akun admin)

PERSIAPAN DAN SET UP PROYEK

Jalankan Web Server = Xampp

Buka cmd buat folder Laravel
composer create-project laravel/laravel nama-anda_JWP_MUK

```
  nesbot/carbon ..... DONE
  nunomaduro/collision ..... DONE
  nunomaduro/termwind ..... DONE

80 packages you are using are looking for funding.
Use the `composer fund` command to find out more!
> @php artisan vendor:publish --tag=laravel-assets --ansi --force

   INFO  No publishable resources for tag [laravel-assets].

No security vulnerability advisories found.
> @php artisan key:generate --ansi

   INFO  Application key set successfully.

> @php -r "file_exists('database/database.sqlite') || touch('database/database.sqlite');"
> @php artisan migrate --graceful --ansi

   INFO  Preparing database.

  Creating migration table ..... 18.03ms DONE

   INFO  Running migrations.

  0001_01_01_000000_create_users_table ..... 28.03ms DONE
  0001_01_01_000001_create_cache_table ..... 18.85ms DONE
  0001_01_01_000002_create_jobs_table ..... 29.78ms DONE


C:\Users\Dinda>
```

Buka folder Laravel “ code .”

```
C:\Users\Dinda>cd pesanmakan

C:\Users\Dinda\pesanmakan>code .
```

Konfigurasi nama database pada file .env dengan nama database db_pemesanan_makanan (atau pesanmakan)

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=pesanmakan
DB_USERNAME=root
DB_PASSWORD=
```

Database & Model (Migration & Seeder)

Buat model, migration, dan controller sekaligus lewat terminal:
php artisan make:model Food -mcr
php artisan make:model Order -mcr
php artisan make:model OrderDetail -m

```
PS C:\Users\Dinda\pesanmakan> php artisan make:model Food -mcr

  INFO  Model [C:\Users\Dinda\pesanmakan\app\Models\Food.php] created successfully.

  INFO  Migration [C:\Users\Dinda\pesanmakan\database\migrations\2026_08_19_010143_create_food_table.php] created successfully.

  INFO  Controller [C:\Users\Dinda\pesanmakan\app\Http\Controllers\FoodController.php] created successfully.

PS C:\Users\Dinda\pesanmakan> php artisan make:model Order -mcr

  INFO  Model [C:\Users\Dinda\pesanmakan\app\Models\Order.php] created successfully.

  INFO  Migration [C:\Users\Dinda\pesanmakan\database\migrations\2026_08_19_010156_create_orders_table.php] created successfully.

  INFO  Controller [C:\Users\Dinda\pesanmakan\app\Http\Controllers\OrderController.php] created successfully.

PS C:\Users\Dinda\pesanmakan> php artisan make:model Orderdetail -mcr

  INFO  Model [C:\Users\Dinda\pesanmakan\app\Models\Orderdetail.php] created successfully.

  INFO  Migration [C:\Users\Dinda\pesanmakan\database\migrations\2026_08_19_010213_create_orderdetails_table.php] created successfully.

  INFO  Controller [C:\Users\Dinda\pesanmakan\app\Http\Controllers\OrderdetailController.php] created successfully.

PS C:\Users\Dinda\pesanmakan>
```

MIGRATION TABLE foods
Buka file database/migrations/xxxx_xx_xx_xxxxxx_create_food_table.php
(PENTING: Pastikan nama tabel di Schema::create adalah 'foods' dengan huruf s, agar relasi foreign key di tabel order_details tidak error)
Ganti sesuai dengan soal:

```php
public function up(): void
{
    Schema::create('foods', function (Blueprint $table) {
        $table->id(); // Primary Key
        $table->string('name');
        $table->enum('category', ['Makanan', 'Minuman', 'Cemilan']);
        $table->integer('price');
        $table->text('description');
        $table->string('image')->nullable(); // Boleh kosong/null
        $table->timestamps(); // created_at & updated_at
    });
}
```

Migration table orders
Buka file database/migrations/xxxx_xx_xx_xxxxxx_create_orders_table.php

```php
public function up(): void
{
    Schema::create('orders', function (Blueprint $table) {
        $table->id(); // Primary Key
        $table->string('customer_name');
        $table->string('table_number'); // Bisa Integer/Varchar (disarankan string/varchar)
        $table->integer('total_price');
        $table->string('status')->default('Pending');
        $table->timestamps();
    });
}
```

Migration table order_details

Migration Tabel order_details (Detail Transaksi + Foreign Key)
File ini berada di folder database/migrations/xxxx_xx_xx_xxxxxx_create_order_details_table.php

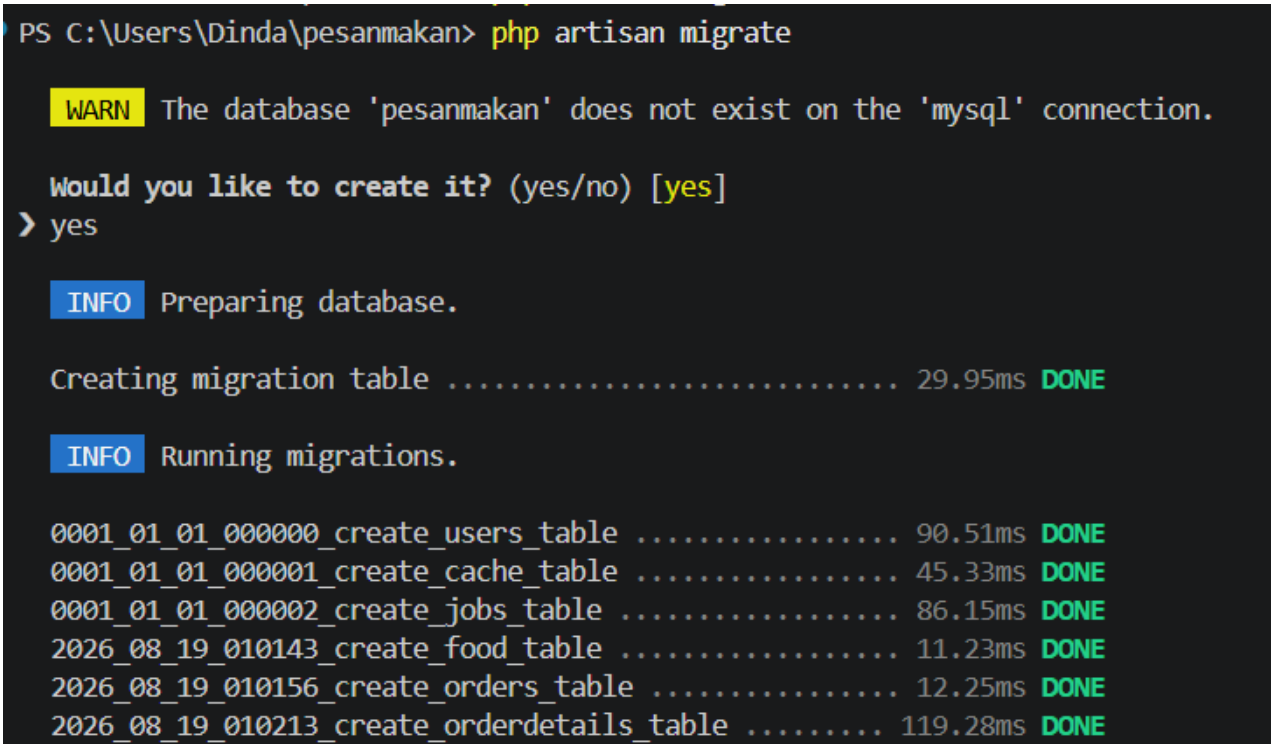
```php
public function up(): void
{
    Schema::create('order_details', function (Blueprint $table) {
        $table->id();

        // Foreign Key ke tabel orders
        $table->foreignId('order_id')
              ->constrained('orders')
              ->onDelete('cascade');

        // Foreign Key ke tabel foods
        $table->foreignId('food_id')
              ->constrained('foods')
              ->onDelete('cascade');

        $table->integer('quantity');
        $table->integer('subtotal');
        $table->timestamps();
    });
}
```

Buka terminal
jalankan php artisan migrate



Seeder digunakan untuk menginput minimal 5 data menu awal ke database secara otomatis.
Buat File Seeder via Terminal Jalankan perintah berikut di terminal VS Code:
php artisan make:seeder FoodSeeder

```php
<?php

namespace Database\Seeders;

use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;
use Illuminate\Support\Facades\DB; // <-- TAMBAHKAN BARIS INI

class FoodSeeder extends Seeder
{
    /**
     * Run the database seeds.
     */
    public function run(): void
    {
        DB::table('foods')->insert([
            [
                'name' => 'Nasi Goreng Spesial',
                'category' => 'Makanan',
                'price' => 25000,
                'description' => 'Nasi goreng dengan telur, ayam suwir, dan kerupuk.',
                'image' => null,
                'created_at' => now(),
                'updated_at' => now(),
            ],
            [
                'name' => 'Mie Goreng Seafood',
                'category' => 'Makanan',
                'price' => 28000,
                'description' => 'Mie goreng pedas dengan udang dan cumi.',
                'image' => null,
                'created_at' => now(),
                'updated_at' => now(),
            ],
            [
                'name' => 'Es Teh Manis',
                'category' => 'Minuman',
                'price' => 5000,
                'description' => 'Es teh melati segar.',
                'image' => null,
                'created_at' => now(),
                'updated_at' => now(),
            ],
            [
                'name' => 'Jus Alpukat',
                'category' => 'Minuman',
                'price' => 15000,
                'description' => 'Jus alpukat murni dengan susu cokelat.',
```

Daftarkan Seeder di database/seeders/DatabaseSeeder.php Buka file DatabaseSeeder.php, lalu tambahkan pemanggilan FoodSeeder di dalam method run():

```php
<?php

namespace Database\Seeders;

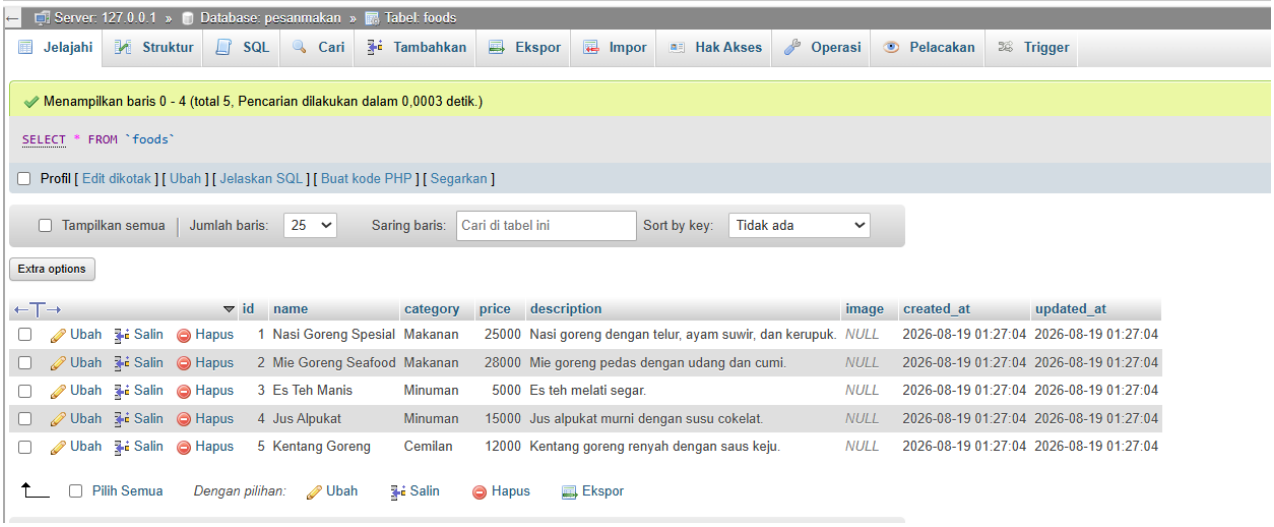
use App\Models\User;
use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;
use Illuminate\Support\Facades\Hash;

class DatabaseSeeder extends Seeder
{
    use WithoutModelEvents;

    /**
     * Seed the application's database.
     */
    public function run(): void
    {
        // 1. Buat User Admin Default
        User::create([
            'name' => 'Admin Toko',
            'email' => 'admin@gmail.com',
            'password' => Hash::make('password123'),
        ]);

        // 2. Panggil Seeder Makanan
        $this->call([
            FoodSeeder::class,
        ]);
    }
}
```

JALANKAN PERINTAH
php artisan migrate:fresh --seed
(Catatan: Wajib gunakan dua strip --seed bukan satu strip)



MODEL

Setup Relasi Eloquent Model

Buka masing-masing file model di folder app/Models/

1. app/Models/Food.php

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Food extends Model
{
    use HasFactory;

    protected $table = 'foods';
    protected $guarded = ['id'];
}
```

2. app/Models/Order.php

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Order extends Model
{
    use HasFactory;

    protected $guarded = ['id'];

    // Relasi: 1 Pesanan punya banyak detail item
    public function orderDetails()
    {
        return $this->hasMany(OrderDetail::class, 'order_id');
    }
}
```

3. app/Models/OrderDetail.php (Gunakan protected $guarded = ['id']; dan nama class OrderDetail)

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class OrderDetail extends Model
{
    use HasFactory;

    protected $table = 'order_details';
    protected $guarded = ['id'];

    // Relasi balik ke model Food
    public function food()
    {
        return $this->belongsTo(Food::class, 'food_id');
    }

    // Relasi balik ke model Order
    public function order()
    {
        return $this->belongsTo(Order::class, 'order_id');
    }
}
```